# Lecture Notes

*Basic Algebraic Structures*

Appendix - *Integers*

Zeki Can Özkanlı

19 February 2024

METU Mathematics ‘22

# CONTENTS

**1 APPENDIX I - Integers** **3**

1.1 Divisibility . . . . . 4

1.2 Prime Factors . . . . . 5

1.3 Congruence of Integers . . . . . 7

CHAPTER

# ONE

# APPENDIX I - INTEGERS

## 1.1 Divisibility

In this chapter, our main goal is to obtain division algorithm. To achieve this we need an important consequence of the induction postulate, known as the Well Ordering Theorem.

**Lemma 1.1 (Least Positive Integer) .**
The integer 1 is the least positive integer.

**Theorem 1.1 (The Well Ordering Principle) .**
Every nonempty set $S$ of positive integers contains a least element. That is, there is an element $m \in S$ such that $m \leq x$ for all $x \in S$.

*Proof.* Let $S$ be a nonempty set of positive integers. If $1 \in S$, then $1 \leq x$ for all $x \in S$, by Lemma 1.1.

Consider now the case where $1 \notin S$, and let $L$ be the set of all positive integers $p$ such that $p < x$ for all $x \in S$. That is,

$$L = \{p \in \mathbb{Z}^+ \mid p < x,\ \forall x \in S\}$$

Since $1 \notin S$, Lemma 1.1 assures us that $1 \in L$. We shall show that there is a positive integer $p_0$ such that $p_0 \in L$ and $p_0 + 1 \notin L$. Suppose this is not the case. Then we have that $p \in L$ implies $p + 1 \in L$ (since $(\neg a \vee b) \iff (a \implies b)$). Now, since $1 \in L$, and $p \in L$ implies $p + 1 \in L$, then by Induction Postulate $L = \mathbb{Z}^+$. But this contradicts the fact that $S$ is nonempty (note that $L \cap S =$, by definition of $L$). Therefore there is a $p_0$ such that $p_0 \in L$ and $p_0 + 1 \notin L$. Now, we must show that $p_0 + 1 \in S$. We have $p_0 < x$ for all $x \in S$, hence, $p_0 + 1 \leq x$ for all $x \in S$. If $p_0 + 1 < x$ were always true, then $p_0 + 1$ would be in $L$. Hence, $p_0 + 1 = x$ for some $x \in S$ and $m = p_0 + 1$ is the required least element in $S$. ■

**Def 1.1 (Divisor) .**
Let $a$ and $b$ be integers. We say that $a$ **divides** $b$ if there is an integer $c$ such that $b = ac$. If $a$ divides $b$, we write $a|b$.

**Theorem 1.2 .**
The only divisors of 1 are 1 and -1.

*Proof.* Suppose $a$ is a divisor of 1. Then $1 = ac$ for some integer $c$. The equation $1 = ac$ requires $a \neq 0$, so either $a \in \mathbb{Z}^+$ or $-a \in \mathbb{Z}^-$. Consider first the case where $a \in \mathbb{Z}^+$. Then this requires $c \in \mathbb{Z}^+$. So we have $1 \leq a$ and $1 \leq c$, by Lemma 1.1. Then,

$$\begin{aligned} 1 < a &\implies 1 \cdot c < a \cdot c \\ &\implies c < 1 \qquad\qquad \text{(since } ac = 1\text{)}. \end{aligned}$$

But this contradicts with that $1 \leq c$. Thus, $1 = a$ is the only possibility when $a \in \mathbb{Z}^+$.

Similar arguments can be done for $-a \in Z^-$.

Therefore, combining the cases where $a \in \mathbb{Z}^+$ and where $-a \in \mathbb{Z}^-$, we have shown that either $a = 1$ or $a = -1$ if $a$ is a divisor of 1. ■

**Theorem 1.3 (The Division Algorithm) .**
Let $a$ and $b$ be integers with $b > 0$. Then there exist unique integers $q$ and $r$ such that

$$a = bq + r \text{ with } 0 \leq r < b.$$

*Proof.* (Existence). Let $S$ be the set of all integers $x$ that can be written in the form $x = a - bn$ for $n \in \mathbb{Z}$, and let $S'$ denote the set of all nonnegative integers in $S$. Then the set $S'$ is nonempty. If $0 \in S'$, we have $a - bq = 0$ for some $q$, and hence $a = bq + 0$. If $0 \notin S'$, then by Well Ordering Principle (1.1) $S'$ contains a least element $r = a - bq$, and hence $a = bq + r$, where $r > 0$. Now,

$$\begin{aligned} r - b &= a - bq - b \\ &= a - b(q + 1), \end{aligned}$$

and hence $r - b \in S$. Since $r$ is the least element in $S'$ and $r - b < r$, then $r - b$ must be negative. That is, $r - b < 0$, hence $r < b$. Therefore, there exist such $q$ and $r$.

(Uniqueness). Suppose that $a = bq_1 + r_1$ and that $a = bq_2 + r_2$, where $0 \leq r_1 < b$ and $0 \leq r_1 < b$. Without loss of generality, we may assume that $r_1 \leq r_2$. This means, $0 \leq r_2 - r_1 \leq r_2 < b$. However, we also have

$$\begin{aligned} 0 \leq r_2 - r_1 &= (a - bq_2) - (a - bq_1) \\ &= b(q_1 - q_2). \end{aligned}$$

That is, $r_2 - r_1$ is nonnegative multiple of $b$ that is less than $b$. Thus, $r_2 - r_1 = 0$, and hence $r_1 = r_2$. Also, it follow that $bq_1 = bq_2$, where $b > 0$. This implies that $q_1 = q_2$. Therefore, we have shown that $r_1 = r_2$ and that $q_1 = q_2$, and this shows that $r$ and $q$ are unique. Therefore the theorem. ■

**Theorem 1.4 (The Strengthened Division Algorithm) .**
Let $a$ and $b$ be integers with $b > 0$. Then there exist unique integers $q$ and $r$ such that

$$a = bq + r \text{ with } 0 \leq r < |b|.$$

*Proof.* Left as an exercise. ■

## 1.2 Prime Factors

**Def 1.2 (Greatest Common Divisor (GCD)) .**
An integer $d$ is a **greatest common divisor** of $a$ and $b$ if the following conditions are hold:

1. $d \in \mathbb{Z}^+$
2. $d \mid a$ and $d \mid b$
3. $(c \mid a) \wedge (c \mid b) \implies c \mid d$.

**Theorem 1.5 .**
Let $a$ and $b$ be integers, at least one of them is not 0. Then there exists a unique greatest common divisor $d$ of $a$ and $b$. Moreover, $d$ can be written as a linear combination of $a$ and $b$, that is,

$$d = am + bn$$

for $m, n \in \mathbb{Z}$, and $d$ is the smallest positive integer that can be written in this form.

*Proof.* (Existence). Let $a$ and $b$ be integers, at least one of them is not 0. If $b = 0$, then $a \neq 0$, and hence $|a| > 0$. It is obvious that $d = |a|$ is a gcd of $a$ and $b$ in this case, and either $d = a \cdot (1) + b \cdot (0)$ or $d = a \cdot (-1) + b \cdot (0)$.

Suppose now that $b \neq 0$. Consider the set $S$ of all integers that can be written in the form $ax + by$ for some integers $x$ and $y$, and let $S^+$ be the set of all positive integers in $S$. The set $S$ contains $b = a \cdot (0) + b \cdot (1)$ and $-b = a \cdot (0) + b \cdot (-1)$, hence $S^+$ is not empty. Then by the Well Ordering Principle 1.1, $S^+$ has a least element $d$,

$$d = am + bn.$$

We have $d$ positive, and we shall show that $d$ is a gcd of $a$ and $b$. By the division algorithm 1.3, there are integers $q$ and $r$ such that

$$a = dq + r \text{ with } 0 \leq r < d.$$

Then,

$$\begin{aligned} r &= a - dq \\ &= a - (am + bn)q \\ &= a(1 - mq) + b(-nq). \end{aligned}$$

Thus, $r$ is in S, and $0 \leq r < d$. By choice of $d$ as the least element in $S^+$, $r = 0$, and hence $d|a$. Similarly, it can be shown that $d|b$. Now, suppose that $c|a$ and $c|b$. Then $a = ch$ and $b = ck$, for some integers $h$ and $k$. Hence,

$$\begin{aligned} d &= am + bn \\ &= chm + ckn \\ &= c(hm + kn), \end{aligned}$$

which shows that $c|d$. Hence, by definition 1.2, $d = am + bn$ is the greatest common divisor of $a$ and $b$. And from the choice of $d$ as least element of $S^+$ that $d$ is the smallest positive integer that can be written in this form.

(Uniqueness). Left as an exercise. ■

**Theorem 1.6 (The Euclidean Algorithm) .**
Let $a$ and $b$ be nonnegative integers. Then $gcd(a, b) = r_n$ is the last nonzero remainder after the following successive process:

$$\begin{aligned} a &= bq_0 + r_1, & 0 \leq r_1 < b \\ b &= r_1 q_1 + r_2, & 0 \leq r_2 < r_1 \\ r_1 &= r_2 q_2 + r_3, & 0 \leq r_3 < r_2 \\ &\vdots \\ r_k &= r_{k+1} q_{k+1} + r_{k+2} & 0 \leq r_{k+2} < r_{k+1}. \end{aligned}$$

*Proof.* ? - Why we always end up with 0, but not the other nonnegative integers. ■

**Def 1.3 (Coprime) .**
Two integers $a$ and $b$ are called **coprime** or **relatively prime** if their greatest common divisor is 1.

**Theorem 1.7 .**
If $a$ and $b$ are relatively prime and $a \mid bc$, then $a \mid c$.

*Proof.* Suppose that $gcd(a, b) = 1$ and $a|bc$. Since $gcd(a, b) = 1$, there exist integers $m$ and $n$ such that $1 = am + bn$, by theorem 1.5. And since $a|bc$, then by

definition there exists an integer $q$ such that $bc = aq$. Now,

$$\begin{aligned} 1 = am + bn &\implies c = amc + bnc \\ &\implies c = amc + aqn \text{ since } bc = aq \\ &\implies c = a(mc + qn) \\ &\implies a|c. \end{aligned}$$

Therefore the theorem. ■

**Def 1.4 (Prime Integer) .**
An integer $p$ is called **prime integer** if $p > 1$ and the only divisors of $p$ are $\pm 1$ and $\pm p$.

**Remark .**
Note that the condition $p > 1$ makes $p$ positive and ensures that $p \neq 1$. The exclusion of 1 from the set of primes makes possible the statement of Unique Factorization Theorem.

**Lemma 1.2 (Euclid's Lemma) .**
If $p$ is a prime and $p|ab$, then either $p|a$ or $p|b$.

*Proof.* Suppose that $p$ is a prime and $p|ab$. If $p|a$, then there is nothing to show. Suppose then that $p \not| a$. This implies that $gcd(p, a) = 1$, since the only positive divisors of $p$ are 1 and $p$. Then by theorem 1.7, $p|b$. Therefore the theorem. ■

**Corollary 1.7.1 .**
If $p$ is a prime and $p|(a_1 a_2 \dots a_n)$, then $p$ divides some $a_i$.

**Theorem 1.8 (Unique Factorization Theorem) .**
Every positive integer $n$ either is 1 or can be expressed as a product of prime integers, and this factorization is unique, except for the order of the factors.

*Proof.* (Existence). Let $P_n$ be the statement that either $n = 1$ or $n$ can be expressed as a product of primes. We shall prove that the statement $P_n$ is true for all $n \in Z^+$ by the Second Principle of Finite Induction. $P_1$ is trivially true. Suppose that $P_m$ is true for all positive integers $m < k$. If $k$ is a prime, then $k$ is a product with one prime factor, and hence $P_k$ is true. Suppose $k$ is not a prime. Then $k = ab$, where neither $a$ or $b$ is 1. Therefore, $1 < a < k$ and $1 < b < k$. By the induction hypothesis, $P_a$ and $P_b$ is true. That is, $a = p_1 p_2 \dots p_r$ and $b = q_1 q_2 \dots q_r$, for primes $p_i$ and $q_i$. Then,

$$\begin{aligned} k &= ab \\ &= p_1 p_2 \dots p_r \cdot q_1 q_2 \dots q_r, \end{aligned}$$

and $k$ is thereby expressed as a product of primes. thus $P_k$ is true, and therefore $P_n$ is true for all positive integers $n$. (Uniqueness). Suppose that $n = p_1 p_2 \dots p_t$ and $n = q_1 q_2 \dots q_v$ are factorizations of $n$ as products of prime factors $p_i$ and $q_j$. Then,

$$p_1 p_2 \dots p_t = q_1 q_2 \dots q_v,$$

So $p_1|(q_1 q_2 \dots q_v)$. By corollary 1.7.1, $p_1|q_j$ for some $j$, and without loss of generality we may assume $j = 1$. However, $p_1$ and $q_1$ are primes, so $p_1|q_1$ implies $q_1 = p_1$. This gives us,

$$p_1 p_2 \dots p_t = p_1 q_2 \dots q_v,$$

and therefore,

$$p_2 \dots p_t = q_2 \dots q_v.$$

By repeating with this process we have finally obtain,

$$p_t = q_t \dots q_v.$$

Since the only positive factors of $p_t$ are 1 and $p_t$, and since $q_j$ is prime, this means that there must be only one $q_j$ on the right-hand side. Hence $v = t$ and $q_t = p_t$. This completes the proof. ■

**Theorem 1.9 (Euclid's Theorem on Primes) .**
There are infinitely many primes.

*Proof.* We shall give the proof by contradiction. Suppose that there are a finite number $n$ of primes, $p_1, p_2, \dots p_n$. Now, consider the integer $m = p_1 p_2 \dots p_n + 1$. It is clear that the remainder in the division of $m$ by any prime $p_i$ is 1, so each $p_i$ is not a factor of $m$. Thus, there are two possibilities, either $m$ is itself a prime, or it has a prime factor that is different from every one of the $p_i$. In either case, we have a prime integer that was not in the list. Therefore, there are more than $n$ primes and this completes the proof. ■

## 1.3 Congruence of Integers

**Def 1.5 .**
Let $n$ be a positive integer, $n > 1$. For integers $x$ and $y$, $x$ is said to be **congruent** to $y$ **modulo** $n$ if $x - y$ is a multiple of $n$, and we write

$$x \equiv y \pmod{n}.$$

# BIBLIOGRAPHY